Finding particular solutions to non-homogeneous differential equations

Particular solutions to non-homogeneous differential equations $F(D)y = f(x)$ is

$$\boxed{y = \frac{1}{F(D)} f(x)}$$

free from arbitrary constants

$$D = \frac{d}{dx} \qquad \frac{1}{D} f(x) = \int f(x)\,dx$$

The operator D denotes differentiation

D^{-1} **is the inverse differential operator**

Result 1

$$\frac{1}{D} f(x) = \int f(x)\,dx$$

We need to show that $$D\left\{\int f(x)\,dx\right\} = f(x)$$

$$D\left\{\int f(x)\,dx\right\} = \frac{d}{dx}\left\{\int f(x)\,dx\right\} = f(x)$$

$$\int f(x)\,dx = \frac{1}{D} f(x)$$

Rule II

$$f(x) = e^{ax}$$ where a is any constant.

$$\frac{1}{F(D)} e^{ax} = \frac{1}{F(a)} e^{ax} \text{ provided } F(a) \neq 0$$

$$F(D)\left\{\frac{1}{F(a)} e^{ax}\right\} = \frac{1}{F(a)}\left\{F(D) e^{ax}\right\}$$

$$= \frac{1}{F(a)} F(a) e^{ax}$$

$$\therefore \boxed{\frac{1}{F(D)} e^{ax} = \frac{1}{F(a)} e^{ax}} \quad F(a) \neq 0$$

Before we discuss the case $F(a) = 0$ will consider Result III.

$F(a) = 0$ වන අවස්ථාව විස්තර කිරීමට ප්‍රථම අප ප්‍රතිඵලය III සලකමු.

Result III

Let $v(x)$ **be a function of** x **only. Then**

$$\frac{1}{F(D)} e^{ax} v(x) = e^{ax} \frac{1}{F(a)} v(x)$$

Using exponential shift rule

$$F(D) e^{ax} u(x) = e^{ax} \underbrace{F(D+a) u(x)}_{v(x)} \qquad \rightarrow (1)$$

Let $$v(x) = F(D+a) u(x)$$

$$\therefore \; u(x) = \frac{1}{F(D+a)} v(x)$$

Substitute to Eq (1)

$$F(D)\, e^{ax} \frac{1}{F(D+a)} \;\; v(x) \;=\; e^{ax} F(D+a) \frac{1}{F(D+a)}\, v(x)$$

$$F(D)\, e^{ax} \frac{1}{F(D+a)} \;\; v(x) = e^{ax}\, v(x)$$

$$\frac{1}{F(D)}\, e^{ax} \;\; v(x) = e^{ax} \frac{1}{F(D+a)}\, v(x)$$

Example 1

$$\frac{1}{\left(D^2 - D - 2\right)} e^{-x} = \frac{1}{\left(D^2 - D - 2\right)} e^{-x} \cdot \underbrace{1}_{V(x)}$$

$$= e^{-x} \frac{1}{\left(D - 1\right)^2 - \left(D - 1\right) - 2} \cdot 1$$

$$= e^{-x} \frac{1}{D^2 - 3D} \cdot 1$$

$$= e^{-x} \frac{1}{D\left(D - 3\right)} \cdot e^{0 \cdot x}$$

$$= e^{-x} \cdot \ e^{0 \cdot x} \frac{1}{D\left(0 - 3\right)}$$

$$= \ e^{-x} \frac{1}{-3D} \left(1\right)$$

$$= -\frac{1}{3} \ e^{-x} \ x$$

Now we consider the value of $\dfrac{1}{F(D)} e^{a x}$ when $F(a) = 0$

If $F(a) = 0$ then $(D - a)$ must be a factor of $F(D)$

$$F(D) = (D - a)^p \, \phi(D) \qquad \phi(a) \neq 0$$

දැන් අප මුල් කොටසේ $\dfrac{1}{F(D)} e^{a x}$ හි අගය $F(a) = 0$ අවස්ථාවේදි සලකා බලමු.

$F(a) = 0$ නම් $(D - a)$, $F(D)$ සාධකයකි.

$F(D) = (D - a)^p \, \phi(D)$ මෙහි යනු ගණය p වන

$F(D)$ හි මූලයකි. මෙහි $\phi(a) \neq 0$ වේ.

$$\frac{1}{F(D)} e^{ax} = \frac{1}{(D-a)^p \ \phi(D)} e^{ax}$$

$$= \frac{1}{(D-a)^p} \frac{e^{ax}}{\phi(a)} .1$$

$$= \frac{1}{\phi(a)} \frac{1}{(D-a)^p} \ e^{ax} \{1\}$$

$$= \frac{1}{\phi(a)} \ e^{ax} \ \frac{1}{(D+a-a)^p} \{1\}$$

$$= \frac{1}{\phi(a)} \ e^{ax} \ \frac{x^p}{D^p}$$

$$\frac{1}{F(D)} e^{ax} = \frac{1}{\phi(a)} \ e^{ax} \ \frac{x^p}{p\,!}$$

Result IV $f(x) = \sin ax$ **or** $f(x) = \cos ax$

Put $D^2 = -a^2$

$D^3 = D^2 D = -a^2 D$ $D^4 = \left(D^2\right)^2 = \left(-a^2\right)^2 = a^4$

$$\frac{1}{F\left(D^2\right)} \sin(ax) = \frac{1}{F\left(-a^2\right)} \sin(ax) \quad ; \; F\left(-a^2\right) \neq 0$$

$$\frac{1}{F\left(D^2\right)} \cos(ax) = \frac{1}{F\left(-a^2\right)} \cos(ax) \quad ; \; F\left(-a^2\right) \neq 0$$

Example 2

(*i*) $$\left(D^2 + 2\right)y = \cos ax$$

$$y_p = \frac{1}{\left(D^2 + 2\right)} \cos ax$$

$$y_p = \frac{1}{(-a)^2 + 2}$$

(*ii*) $$\left(D^2 + a^2\right)y = \sin ax$$

$$y_p = \frac{1}{D^2 + a^2} \sin ax$$

$$\boxed{e^{iax} = \cos ax + i \sin ax}$$

$$y_p = \textbf{Imaginary part of } \frac{1}{D^2 + a^2} e^{iax}$$

$$y_p = \textbf{Imaginary part of } \frac{1}{D^2 + a^2} e^{iax}$$

$$\boxed{\frac{1}{F(D)} e^{ax} v(x) = e^{ax} \frac{1}{F(D+a)} v(x)}$$

$$y_p = \frac{1}{D^2 + a^2} e^{iax} .1 = e^{iax} \frac{1}{(D+ia)^2 + a^2} 1$$

$$= e^{iax} \frac{1}{D^2 + 2iaD} e^{0.x} = e^{iax} \frac{1}{D} \frac{1}{D+2ia} e^{0.x}$$

$$= e^{iax} \frac{1}{D} \frac{1}{0+2ia} e^{0.x} = \frac{1}{2ia} \frac{1}{D} 1 = \frac{1}{2ia} e^{iax} x = \frac{x(\cos ax + i \sin ax)}{2ia}$$

$$= \textbf{Imaginary part of } \frac{x(-i \cos ax + \sin ax)}{2a} = \frac{x \sin ax}{2a} - \frac{i \cos ax}{2a}$$

$$y_p = -\frac{x}{2a} \cos x$$

(ii) $\left(D^2 + 3D - 4\right)y = \mathbf{sin}\ 2x$

$$y_p = \frac{1}{\left(D^2 + 3D - 4\right)} \mathbf{sin}\ 2x$$

$$= \frac{1}{\left(D + 4\right)\left(D - 1\right)} \mathbf{sin}\ 2x$$

$$= \frac{\left(D - 4\right)\left(D + 1\right)}{\left(D^2 - 16\right)\left(D^2 - 1\right)} \mathbf{sin}\ 2x$$

$$= \left(D - 4\right)\left(D + 1\right) \frac{1}{\left(-2^2 - 16\right)\left(-2^2 - 1\right)} \mathbf{sin}\ 2x$$

$$= \frac{1}{100} \left(D^2 - 3D - 4\right) \mathbf{sin}\ 2x$$

$$= \frac{1}{100} \left[-4\ \mathbf{sin}\ 2x - 6\ \mathbf{cos}\ 2x - 4\ \mathbf{sin}\ 2x\right]$$

$$= \frac{1}{50} \left[-4\ \mathbf{sin}\ 2x - 3\ \mathbf{cos}\ 2x\right]$$

Result V

$$f(x) = x^m$$

$$\frac{1}{F(D)} x^m = \left(a_0 + a_1 D + a_1 D^2 + \dots + a_1 D^n\right) x^m \quad ; a_0 \neq 0$$

$\frac{1}{F(D)}$ යන්න D හි බලය වැඩිවන පිළිවෙලට ලියා D^m වලින් එහාට ඇති පද ඉවත් කරන්න.

Write $\frac{1}{F(D)}$ in the ascending powers of D up to the power D^m and the terms exceeding D^m going to be zero.

$$D^n x^m = 0 \quad ; n > m$$

$$f(x) = x^m$$

$$\frac{1}{F(D)} x^m = \left(a_0 + a_1 D + a_1 D^2 + \ldots + a_1 D^n\right) x^m \quad ; a_0 \neq 0$$

$$D^n x^m = 0 \qquad ; n > m$$

To find the Particular Integral (PI) of this type the following expansions in the ascending powers of *D* will be useful.

$$\frac{1}{1+D} = 1 - D + D^2 - D^3 + D^4 + \ldots$$

$$\frac{1}{1-D} = 1 + D + D^2 + D^3 + D^4 + \ldots$$

Example 3

$$\left(2D^2 + 2D + 3\right)y = x^2 + 2x - 1$$

$$y = \frac{1}{\left(2D^2 + 2D + 3\right)} x^2 + 2x - 1$$

$$= \frac{1}{3\left[1 + \frac{2}{3}D\left(D+1\right)\right]} x^2 + 2x - 1$$

$$= \frac{1}{3}\left[1 - \frac{2}{3}D\left(D+1\right) + \frac{4}{9}D^2\left(D+1\right)^2 + \ldots\right] x^2 + 2x - 1$$

$$= \frac{1}{3}\left[1 - \frac{2}{3}D^2 - \frac{2}{3}D + \frac{4}{9}D^2 + \ldots\right] x^2 + 2x - 1$$